



AI Evaluation Report

Data Science Expert — Self-Assessment

CANDIDATE

Alex Johnson

FINAL SCORE

14 / 70

weighted points

CORRECT ANSWERS

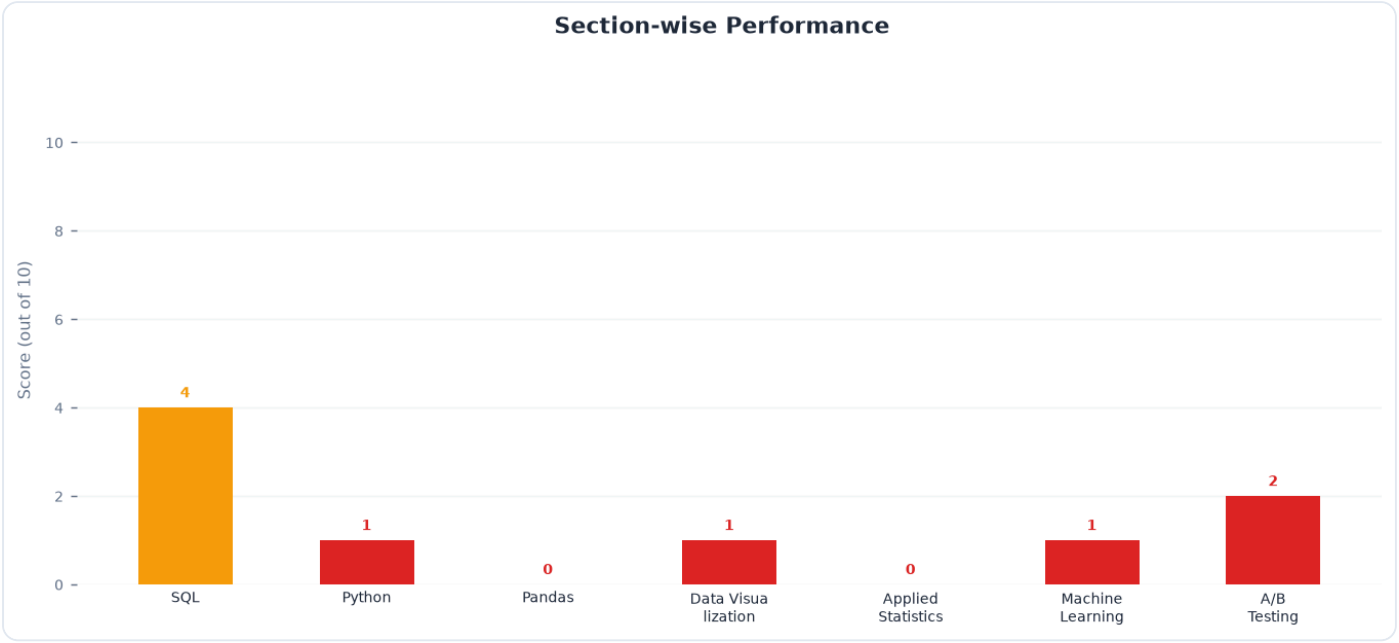
9 / 35

questions

NEEDS IMPROVEMENT

Assessment Date: July 26, 2026 Report Generated: July 26, 2026 at 15:28 UTC Role: Data Science Expert

OVERALL RATING	SUCCESS RATE	WEIGHTED SCORE	TOP SKILL	NEEDS IMMEDIATE ATTENTION
★☆☆☆☆	20.0%	14 / 70	SQL	Pandas



Skill Rating & Benchmark

Section	Your Score	Reference Benchmark (Demo)	Delta	Status
SQL	4 / 10	6.8	-2.8	Needs Work
Python	1 / 10	6.2	-5.2	Critical Gap
Pandas	0 / 10	5.8	-5.8	Critical Gap
Data Visualization	1 / 10	6.0	-5.0	Critical Gap
Applied Statistics	0 / 10	5.5	-5.5	Critical Gap
Machine Learning	1 / 10	5.2	-4.2	Critical Gap
A/B Testing	2 / 10	6.4	-4.4	Critical Gap

Attempt Statistics Breakdown

Section	Total Q	Attempted	Correct	Incorrect	Skipped	Accuracy
SQL	5	5	4	1	0	80%
Python	5	5	1	4	0	20%
Pandas	5	5	0	5	0	0%
Data Visualization	5	5	1	4	0	20%
Applied Statistics	5	5	0	5	0	0%
Machine Learning	5	5	1	4	0	20%
A/B Testing	5	5	2	3	0	40%

Executive Summary

The candidate achieved 9 out of 35 correct answers (25.7% accuracy) with a weighted score of 14 out of 70. This performance indicates significant gaps in foundational Python/Pandas programming, statistical inference, and core machine learning mechanics. To be competitive for data science roles, a structured technical upskilling plan is urgently required.

Key Strengths

- High-Level Methodological Awareness:** Demonstrates familiarity with advanced concepts like sequential A/B testing and descriptor protocols, showing exposure to industry-standard terminology.
- Broad ML Vocabulary:** Recognizes key machine learning frameworks and ensemble methods, providing a baseline vocabulary to build upon.

Top Blind Spots

- **Python & Pandas Foundations:** Gaps in basic syntax execution, including string/set operations, dictionary unpacking order, and crucial Pandas operations such as `transform`, `fillna` mechanics, and MultiIndex sorting.
- **Statistical Inference & A/B Testing:** Misinterpretation of fundamental statistical concepts, including the empirical rule, standard error calculations, p-value and confidence interval interpretations, and sample size drivers.
- **SQL & ML Execution Details:** Misunderstanding database index behaviors during window functions versus correlated subqueries, as well as the mathematical mechanics of L1 regularization and distance-based algorithms.

Actionable Learning Resources

Pandas & Python: Complete the "Pandas" track on Kaggle Learn, focusing specifically on grouping, multi-indexing, and `transform` operations.

✓ Action Item

Statistics & A/B Testing: Read Chapters 2 and 3 of *OpenIntro Statistics* to master hypothesis testing, and practice sample size calculations using Evan Miller's online A/B testing calculators.

✓ Action Item

Machine Learning Theory: Read Chapter 3 (Linear Regression) and Chapter 6 (Regularization) of *An Introduction to Statistical Learning (ISLR)* to understand VIF, Lasso, and feature scaling.

✓ Action Item

SQL Optimization: Work through the "Advanced Select and Joins" section on HackerRank, analyzing execution plans to compare window functions against correlated subqueries.

✓ Action Item